

Glutamine synthetase expression in perinatal spiny mouse liver

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The pronounced increase in the protein/mRNA ratio of ammonia-metabolising enzymes in rat liver in the last prenatal week represents a clear example of a post-transcriptional level of control of gene expression. Both the underlying mechanism, namely an increase in translational efficiency of the mRNA and/or enhanced stability of the protein, and its importance for perinatal adaptation are unknown. We investigated this process in spiny mouse liver, because the comparison of rat and spiny mouse can discriminate adaptively from developmentally regulated processes in the perinatal period. We focused on glutamine synthetase (GS) because of the conveniently small size of its mRNA. Prenatally, GS enzyme activity slowly accumulated to $\approx 1.3 \text{ U} \cdot \text{g}^{-1}$ liver at birth and postnatally more rapidly to $5.5 \text{ U} \cdot \text{g}^{-1}$ at 2 weeks. Both phases of enzyme accumulation obeyed exponential functions. Western-blot analysis showed that changes in GS activity reflected changes in GS protein content. GS mRNA content of the liver was $45 \text{ fmol} \cdot \text{g}^{-1}$ at 2 weeks before birth and slowly declined to $\approx 25 \text{ fmol} \cdot \text{g}^{-1}$ at 2 weeks after birth. The GS protein/mRNA ratio increased 2.5-fold prenatally and sixfold postnatally. Analysis of prenatal and postnatal polysome profiles revealed no evidence of GS mRNA-containing ribonucleoprotein particles. Instead, GS mRNAs were (fully) occupied by 12 ribosomes, indicating regulation at the level of elongation. The kinetics of GS protein accumulation, in conjunction with GS mRNA content, are consistent with an $\approx$ sixfold increase in its rate of synthesis at birth as the result of a corresponding stimulation of the rate of elongation.

Keywords: glutamine synthetase; liver; perinatal adaptation; spiny mouse; translational control.

It is now well established that the developmental appearance of tissue-specific enzymes in the liver shows a perinatal and a preweaning phase of acceleration and that hormones, in particular adrenal and pancreatic hormones, play a major role in these developmental processes [1,2]. Many studies have established that these hormones exert their effects to a large extent at the transcriptional level [3,4]. Nevertheless, we observed in perinatal rat liver that the protein/mRNA ratio of the ammonia-metabolising enzymes carbamoylphosphate synthetase [5], glutamine synthetase (GS) [6] and glutamate dehydrogenase [7] is more than 10-fold lower before than after birth. These perinatal increases in protein/mRNA ratio represent a clear example of a post-transcriptional level of control of gene expression and imply that these proteins accumulate more rapidly as a result of increased translational efficiency or enhanced protein stability. Because sufficiently accurate data for other enzymes of the perinatal cluster [1] are not (yet) available (both mRNA and protein levels are low in this period), it unfortunately remains to be established if the finding is a feature that reflects the onset of birth-preparatory maturation of the liver.

In previous studies, we have compared perinatal development of the spiny mouse, a precocial murinoid species, with that of the rat, a closely related altricial species. This comparison [8–15]

showed that structural development proceeds at a similar rate in both species, but that birth and birth-associated adaptive processes are timed differently. As a result, the first 8–10 postnatal days in the rat are comparable, in developmental terms, with the last 8–10 prenatal days in the spiny mouse. For this reason, rat and spiny mouse are a model system to investigate to what extent perinatal processes have a developmental basis, that is, are related to the time elapsed since fertilisation, or reflect an adaptive response to birth. We have therefore investigated the perinatal changes in GS mRNA and protein content in spiny mouse liver, and have compared these results with the earlier findings in the rat. We focused on the GS gene, because its mRNA is sufficiently small to allow a detailed analysis of its polysomes.

MATERIALS AND METHODS

Animals

Spiny mice (*Acomys cahirinus*) were obtained from our laboratory colony. This colony consists of two inbred lines, each with over 25 generations of brother–sister matings. Animals were housed with a 12-h light and 12 h-dark cycle and permitted *ad libitum* access to water and pelleted rat chow (AM-II diet; Hope Farms, Woerden, the Netherlands). The study was performed in accordance with the Dutch guidelines for the use of experimental animals.

Tissue preparation

Animals were anaesthetised in CO_2/O_2 and killed by decapitation. Fetuses were delivered by caesarian section. Fetal age was

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Abbreviations: GS, glutamine synthetase; RT-PCR, reverse transcriptase PCR.

Enzymes: carbamoylphosphate synthetase (EC 6.3.4.16); glutamine synthetase (EC 6.3.1.2); glutamate dehydrogenase (EC 1.4.1.3).

(Received 16 February 1999; accepted 25 March 1999)

read from a chart relating body weight to gestational age [9]. Birth usually occurs on day 39 of pregnancy. Livers were quickly removed, weighed, frozen in liquid N₂ and stored at -80 °C until assay.

Assays

GS activity was estimated by the γ -glutamyltransferase assay. Briefly, a 10% homogenate was prepared in an ice-cold solution of 150 mM KCl, 1 mM dithiothreitol and 0.1% Triton X-100. γ -Glutamyltransferase activity (1 U = 1 μ mol·min⁻¹) in the supernatant was determined at 37 °C, essentially as described [16]. The protein concentration in the homogenate was determined with the bicinchoninic protein assay (Pierce, Rockford, IL, USA).

Total liver RNA was isolated by the guanidinium isothiocyanate/acid phenol extraction procedure [17], modified to remove residual DNA by precipitation in 2 M LiCl for 18 h at 4 °C. Total RNA concentration in liver was estimated with the orcinol method [18]. Briefly, perchloric acid was added to a 20% liver homogenate in ice-cold water to a final concentration of 0.7 M. RNA in the pellet was hydrolysed for 60 min at 37 °C in 5 vol. 0.3 M KOH. After acidification with an equal volume of ice-cold 0.45 M perchloric acid and centrifugation, ribose was assayed in the extract as described [18].

GS mRNA was estimated by quantitative reverse transcriptase (RT)-PCR [19]. The nucleotide sequence of spiny mouse GS mRNA was determined by sequencing several genomic and cDNA PCR fragments (F. J. van Hemert, W. T. Labruyère & W. H. Lamers, unpublished results). Both GS-specific primers (nt 56–74 and 223–240 of the spiny mouse cDNA) were chosen to span the first intron. Total RNA (0.5 μ g) was reverse-transcribed with 20 units SuperscriptTM II (Life Technologies, Gaithersburg, MD, USA) for 60 min at 42 °C in the presence of 40 pmol of each of the primers and 25 μ g BSA in 75 mM Tris/HCl (pH 9.0), containing 20 mM (NH₄)₂SO₄, 1.25 mM MgCl₂, 0.01% Tween 20 and 100 μ M dNTPs. After an initial denaturation for 2 min at 94 °C, the cDNA was amplified with 0.25 unit *Taq* polymerase (Goldstar; Eurogentec, Seraing, Belgium) for a total of 30 cycles according to a 'touch-down' protocol in which the mixture was denatured for 30 s at 94 °C at the beginning of each cycle, annealed for 30 s at 68 °C (this temperature was lowered 0.5 °C each cycle during the initial 16 cycles), and extended for 45 s at 68 °C. The concentration of GS mRNA in 0.5 μ g total RNA was estimated by titrating 0.08–50 amol of an mRNA copy of spiny mouse GS sequence 1–453, containing a 244-bp insert at position 117, into the RT-PCR assay. The 'competimer' or 'mimic' fragment was quantified by including a known amount of [³⁵S]thio]CTP during large-scale synthesis of the competimer with the RiboMaxTM system (Promega, Madison, WI, USA). Fragments were separated on 1.5% pronarose MS-8 (Hispanagar, Burgos, Spain) in 0.5 $\times$ TBE buffer (0.09 M Tris, 0.89 M H₃BO₃, 0.02 M EDTA, pH 8.0) and quantified with the Eagle Eye II system (Stratagene, La Jolla, CA, USA). GS mRNA in the sample was calculated by plotting the logarithm of the GS/competimer ratio as a function of the logarithm of the competimer concentration [20].

Western-blot analysis

Aliquots of liver protein extract were separated on 10% (w/v) polyacrylamide gels in the presence of SDS and subsequently blotted on to 0.45 μ m poly(vinylidene difluoride) membrane (Immobilon-P; Millipore Corporation, Bedford, MA, USA) as described [7]. After staining with Amido Black to demonstrate protein loading, the membrane was stained for the presence of

GS protein by incubation for 6 h in 10 mM Tris/HCl (pH 8.0), containing 150 mM NaCl, 0.5% Tween 20 and 5% non-fat dried milk powder (Natrion; Nutricia, Zoetermeer, The Netherlands), followed by an overnight incubation in a 1 : 3000 dilution of a mouse monoclonal antibody against GS (clone GS-6; Affiniti Research products, Nottingham, UK). Antibody binding was visualised by the indirect unconjugated peroxidase-antiperoxidase technique, using 3,3'-diaminobenzidine as chromogenic substrate.

Analysis of polysome profiles

Livers were excised and immediately homogenised in 50 mM Hepes (pH 7.8) containing 200 mM sucrose, 70 mM KCl, 10 mM MgCl₂, 5 mM NaCl, 4 mM vanadyl ribonucleoside, 1 mM EGTA, 0.2 mg·mL⁻¹ heparin, 1 mM dithiothreitol, 100 U·mL⁻¹ RNasin [21] and 0.2 mg·mL⁻¹ cycloheximide. After the addition of deoxycholate to a final concentration of 1%, 0.5 mL homogenate ($\approx$ 20 A₂₆₀ units) was loaded on top of an isokinetic 0.5–1.25 M sucrose gradient [22]. After centrifugation for 90 min at 275 000 *g*_{max} (5 °C, Beckman SW40 rotor), the gradient was scanned through the flow-cell of a spectrophotometer and fractionated into 30 fractions of 0.45 mL. The fractions were prepared for dot-blot analysis as described [23]. Standard conditions were used for GS-probe labelling and hybridisation [24]. Bound radioactivity was quantified with a PhosphorImagerTM (Molecular Dynamics).

Statistics

Non-linear regression was used to fit the measured data points. For each fitted curve the standard error and confidence interval of the predicted values was calculated from the residual variance [25]. The time course of GS mRNA content per g of liver and mU GS activity per fmol GS mRNA were determined as the product of GS mRNA content per μ g total RNA and the total RNA content per g of liver, and as the quotient of GS activity and mRNA content per g of liver, respectively. The standard errors for these predictions were determined for each time point according to the rules for accumulation of errors [26]. Briefly, the relative error of a product is the sum of the relative errors of the factors; similarly the relative error of a quotient is the sum of the relative errors of the numerator and the denominator. The *t*-value used for the construction of confidence intervals for these derived functions was based on the lowest number of original observations.

RESULTS

Developmental changes in GS activity

Figure 1A shows that GS activity could already be detected at 23 days of gestation (16 days before birth; 0.7 U·g⁻¹), the youngest age-group investigated. Morphologically, such animals are comparable to rat fetuses at 17 days of gestation (5 days before birth) [8]. GS activity increased slowly with increasing age to a value of $\approx$ 1.3 U·g⁻¹ at birth. After birth, enzyme activity increased rapidly to reach a value of $\approx$ 5.5 U·g⁻¹ at 5 weeks. No differences were observed between male and female animals (not shown). GS activity in adult animals (3–4 months of age) was $\approx$ 20% lower (4.5 ± 0.4 U·g⁻¹). Because of the pronounced difference in enzyme accumulation before and after birth, the prenatal and postnatal developmental changes in GS enzyme activity were separately fitted with the exponential function $y = c + ae^{-kt}$ [27]. Before birth, parameter values were $k = 0.003\cdot\text{day}^{-1}$, $R^2 = 0.50$, and $P = 0.03$ (two-tailed test).

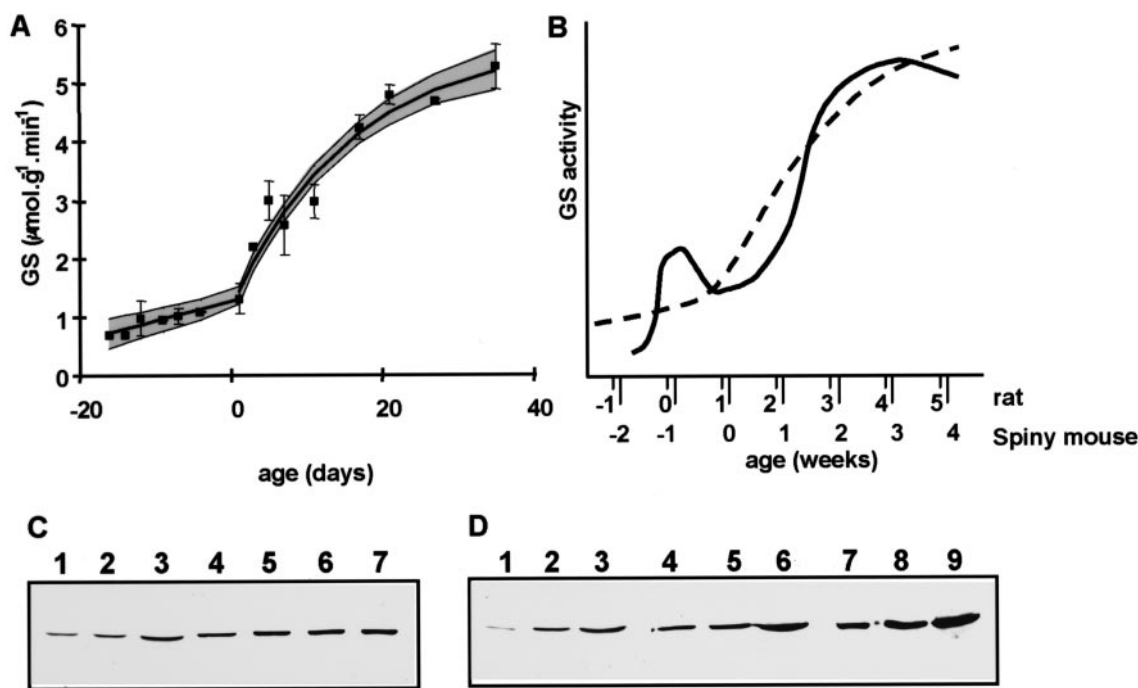
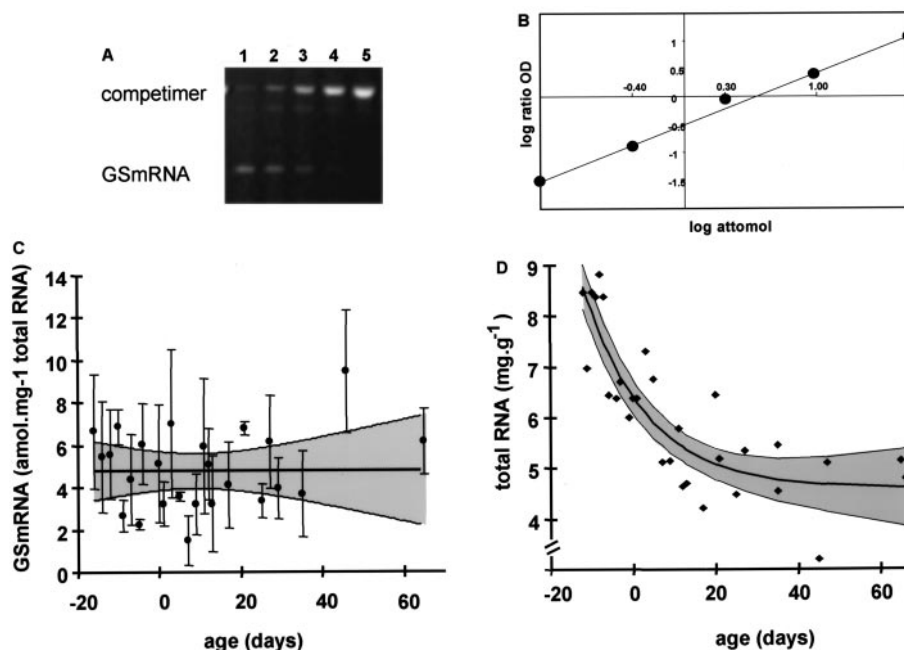


Fig. 1. Developmental changes in GS activity. (A) Developmental changes in GS activity in spiny mouse liver. Enzyme activity was determined by the γ -glutamyltransferase assay. Data represent four animals per time point and are shown as mean $\pm$ SEM. If no error bars are visible, the SEM is smaller than the symbol. The thick line represents the calculated regression line of the data, while its 95% confidence interval is shaded. (B) Comparison of developmental appearance of GS activity in spiny mouse (dashed line) and rat liver (solid line). The position of the age of both species, which is shown on the horizontal axis, is adjusted so that their developmental stage at each point is comparable [8]. (C) and (D) Western-blot analysis of protein extracts from spiny mouse liver at different developmental stages. Protein extracts were prepared from liver at 7 and 3 days before birth, at birth, and at 12, 21 and 45 days after birth and from adult liver. In (C), samples containing identical amounts of GS activity [0.25 mU; 38 (lane 1), 27 (lane 2), 45 (lane 3), 8 (lane 4), and 5 (lanes 5–7) μ g of liver protein, respectively] were separated on a 10% polyacrylamide/SDS gel, blotted on to a poly(vinylidene difluoride) membrane and stained immunochemically for the presence of GS protein (43 kDa). In (D), the effects of different protein loads [4, 20 and 100 μ g protein extract from liver at 3 days before birth (lanes 1–3), 12 days after birth (lanes 4–6) and adult (lanes 7–9)] on immunochemical staining intensity were analysed.

Fig. 2. Determination of GS mRNA by quantitative RT-PCR.

(A) Mixtures of GS mRNA-containing liver extracts (lower band) and fivefold increments of a known amount of [35 S]thioCTP-labelled RNA competitor (upper band) were reverse transcribed and coamplified for 30 cycles. (B) The concentration of GS mRNA in the extract is calculated by determining the regression of the ratio of the concentration of the GS and the competitor bands (y axis) and the original concentration of the competitor in the mixture. At the equivalence point [log ratio absorbance (A) = 0], the concentration of the competitor and GS mRNA are equal. (C) Developmental changes in GS mRNA content per μ g total RNA in spiny mouse liver. Data represent two to three animals per time point and are shown as mean $\pm$ variance or SEM. (D) Developmental changes in total RNA content per gram liver of spiny mouse liver. Total RNA was determined with the orcinol assay. The thick line in (C) and (D) represents the calculated regression line of the data, while its 95% confidence interval is shaded.



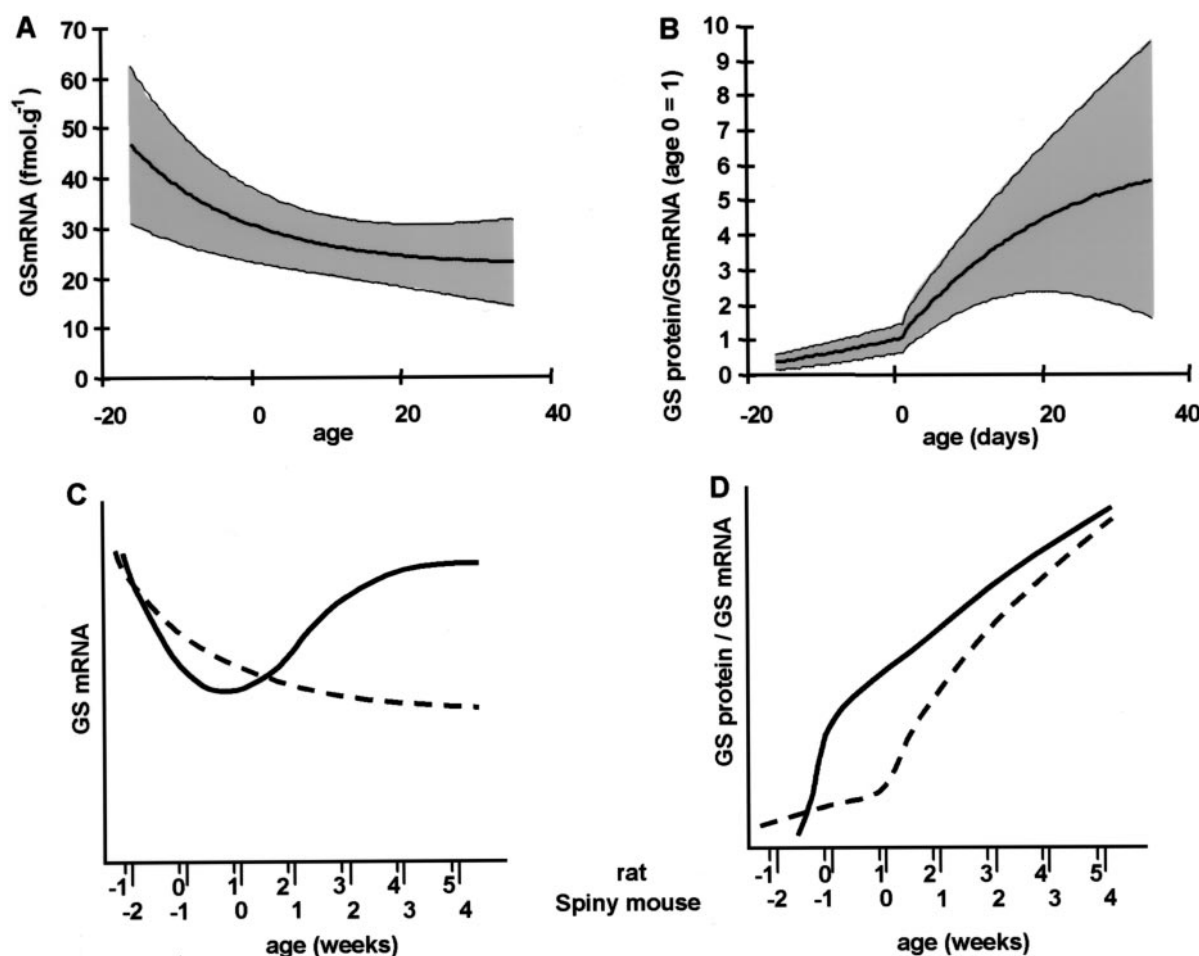


Fig. 3. Developmental changes in GS mRNA. (A) Calculated developmental changes in GS mRNA content per gram of liver (data from Fig. 2C multiplied by those from Fig. 2D). (B) Calculated developmental changes in the GS protein/mRNA ratio in spiny mouse liver (data from Fig. 1A divided by those from Fig. 3A). The thick line in (A) and (B) represents the calculated regression line of the data, while its 95% confidence interval is shaded. (C) and (D) Comparison of the developmental changes in GS mRNA content (C) and GS protein/GS mRNA ratio (D) in spiny mouse (dashed line) and rat (solid line) liver. The position of the age of both species, which is shown on the horizontal axis, is adjusted so that their developmental stage at each point is comparable [8].

After birth, these values were $k = 0.06 \cdot \text{day}^{-1}$, $\text{GS}_{\text{max}} = 6 \text{ U} \cdot \text{g}^{-1}$, $R^2 = 0.91$, and $P < 0.001$. Analysis of the reduction in residual variances showed that a fit to two exponential functions was significantly better than to a single straight line ($F = 7.89$; $\text{df} = 2, 23$; $P = 0.0025$). Figure 1B shows the comparison of the developmental appearance of GS in spiny mouse and rat liver [28]. This comparison of a precocial and an altricial mode of development shows that the precocial spiny mice lacks the typical perinatal rise in enzyme activity seen in the altricial rat, whereas the postnatal increase in enzyme activity in spiny mice coincides with the preweaning increase in enzyme activity in the rat (for a more extensive discussion, see [11,14]).

To assess whether enzymic activity per molecule of GS protein changed during development, equal amounts of enzyme activity were separated on a denaturing polyacrylamide gel, blotted on to poly(vinylidene difluoride) membrane, and stained for the presence of GS protein. Figure 1C shows that the mobility of GS protein did not change, but that the intensity of the GS band increased slightly with development. Because the same activity represented an up to eightfold larger aliquot of extract before than after birth, the effect of varying the protein content per lane was investigated. Figure 1D shows that fivefold increases in protein load (compare lanes 1 and 3, 4 and 6, and 7 and 9) were accompanied by increases in the intensity of the

stained GS signal, but not proportionally. This observation indicated that the amount of GS protein present in the prenatal liver extracts was underestimated relative to that present in the postnatal extracts. We therefore concluded that the amount of GS protein per unit of γ -glutamyltransferase activity did not change appreciably or, at most, increased slightly with development. In any case, developmental changes in GS enzymic activity largely reflected developmental changes in GS protein content.

Developmental changes in GS mRNA content

Developmental changes in hepatic GS mRNA content were determined by competitive RT-PCR. Figure 2A shows the amplification of the GS mRNA-specific fragment (lower band) in the presence of fivefold increments of the competitor. Figure 2B shows the determination of the equivalence point at which the concentration of the competitor (which is known) and GS mRNA are equal (log ratio absorbance = 0). This method was used to determine GS mRNA concentration as a fraction of total liver RNA between 16 days before birth and 2 months after birth (Fig. 2C). Linear regression analysis of the data revealed a slope of 0.014 (t -test: $t = 0.65$; $P = 0.52$), so that hepatic GS mRNA content as a fraction of total RNA can be

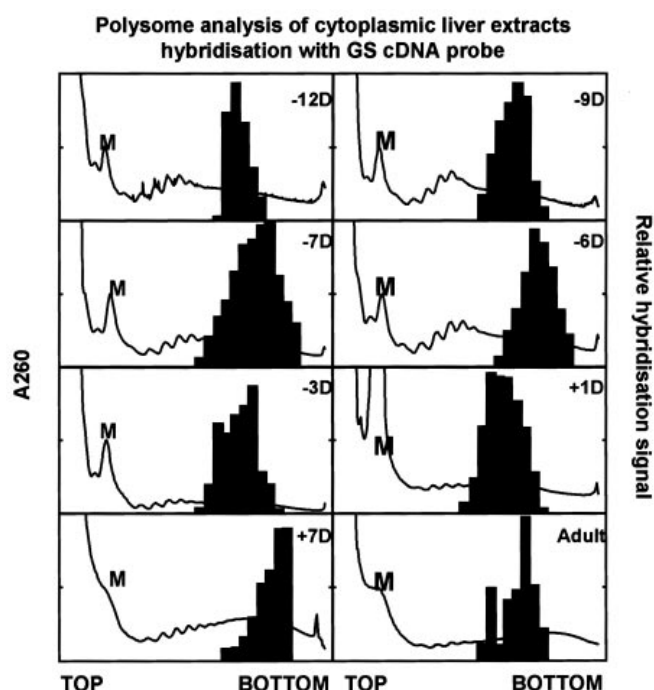


Fig. 4. Ribosomal occupancy of GS mRNA in developing spiny mouse liver. Cytoplasmic extracts were prepared from livers of the indicated ages (D, days). Thirty fractions were obtained after centrifugation (200 000 g_{av} for 90 min at 5 °C) of the extract (0.5 mL or 20 A_{260} units) in an isokinetic sucrose gradient (0.5–1.25 M). The presence of monosomes and polysomes in the fractions was determined by spectrophotometry (A_{260}) and is indicated by a solid line. M, the position of the monosomal peak. The relative abundance of GS mRNA in each fraction (black bars) was determined by dot-blot hybridisation.

considered to be constant with a mean of 4.8 amol per μg total RNA (t -test: $t = 10.5$; $P = 0.000$). To be able to relate GS mRNA to GS protein expression, developmental changes in total RNA content of the liver were determined. Figure 2D shows that at 2 weeks before birth, total RNA content amounted to $\approx 8.5 \text{ mg} \cdot \text{g}^{-1}$ and declined to $\approx 5 \text{ mg} \cdot \text{g}^{-1}$ at 2 weeks after birth. The developmental decline in total RNA content of the liver could be fitted well to an exponential decay function ($y = c + ae^{kt}$; a and $c > 0$, and $k < 0$; $R^2 = 0.73$). In 2-month-old animals, total RNA concentration in the liver had settled at $\approx 4.6 \text{ mg} \cdot \text{g}^{-1}$. The calculated GS mRNA content per g of tissue therefore slowly declined from $\approx 45 \text{ fmol} \cdot \text{g}^{-1}$ at 2 weeks before birth via $\approx 30 \text{ fmol} \cdot \text{g}^{-1}$ at birth to $\approx 25 \text{ fmol} \cdot \text{g}^{-1}$ at 2 weeks after birth (Fig. 3A). Figure 3C shows, for comparison, the developmental changes in hepatic GS mRNA content in both spiny mouse and rat liver [6]. The comparison shows that GS mRNA content in both species declines prenatally, but also that GS mRNA content in the rat increases again postnatally, but not so in the spiny mouse.

Developmental changes in GS protein/mRNA ratio

Under steady-state conditions the protein/mRNA ratio reflects the translatability of the mRNA and the stability of the protein. Figure 3B shows that the GS protein/mRNA ratio in spiny mouse liver increased ≈ 2.5 -fold in the last 2 weeks before birth and five- to six-fold in the first 5 weeks after birth. A comparison of the developmental changes in this ratio between spiny mouse and rat (Fig. 3D) shows that, in both species, the

ratio is low prenatally, increases rapidly in the perinatal period and more gradually during the suckling period.

Developmental changes in ribosomal occupancy of GS mRNA

Changes in translatability of mRNAs are often marked by unmasking of the transcript from stored messenger ribonucleoprotein particles (reviewed in [29]). We therefore analysed the polysome profiles of cytoplasmic extracts of spiny-mouse liver at various ages (Fig. 4). The positions of mRNAs loaded with up to six to eight ribosomes could be identified as separate peaks and were used to calibrate the gradient. GS mRNA was found to be occupied by about 12 ribosomes from the youngest stage investigated (12 days before birth) onward. As an actively translating ribosome occupies ≈ 90 nucleotides [30] and as the reading frame of the spiny mouse GS transcript is 1122 nt in length (F. J. van Hemert, W. T. Labruyère & W. H. Lamers, unpublished work), GS mRNA is always occupied by a maximum load of ribosomes. No evidence of GS mRNA-containing ribonucleoprotein particles was seen.

DISCUSSION

Changes in the protein/mRNA ratio

When the rate of synthesis or degradation changes, macromolecules with a short half-life attain their new steady-state level more quickly than those with a longer half-life [27]. Hence, a change in the protein/mRNA ratio of a gene can have a kinetic basis. However, changes that are due to a kinetic background only should be temporary, so that this explanation can be rejected (see also our arguments in [5]).

We did not observe appreciable developmental changes in the enzymatic activity per GS protein molecule (Fig. 1C,D). Similarly, we did not notice developmental changes in molecular specific activity for genes characterised by a developmental increase in the protein/mRNA ratio in rat liver [7,31]. Furthermore, we did not observe a change in the electrophoretic mobility of the immunoreactive GS protein (Fig. 1C,D). These findings indicate that post-translational modification of GS protein is an unlikely cause of the observed increase in the protein/mRNA ratio.

A permanent change in the protein/mRNA ratio can result from a permanent change in the rate of protein synthesis or degradation of the protein. We have no detailed information about these variables in the perinatal period. However, the findings that GS mRNA concentration in spiny mouse liver does not change appreciably perinatally, whereas GS activity does, and that this accumulation of GS activity is consistent with an exponential function, allow some preliminary, but interesting, calculations. An exponentially inclining accumulation of (GS) protein is found when the rate of protein synthesis increases, whereas the rate of protein degradation does not change [27,32]. In this case, the observed time constant 0.06 day^{-1} corresponds to a half-life of GS protein of 11–12 days and a rate of synthesis of $0.35 \text{ U} \cdot \text{g}^{-1} \cdot \text{day}^{-1}$ after birth [27]. A half-life of 11–12 days of GS protein in spiny mouse liver is about twice that reported for rat liver [33]. Taking into account that the GS mRNA concentration in spiny mouse liver (Fig. 2C) is only $\approx 50\%$ of that in rat (Fig. 3C; [5]) and mouse (W. T. Labruyère and W. H. Lamers, unpublished data) liver, and that total hepatic RNA content and polysome loading respond similarly to birth in both spiny mouse (Figs 2D and 4) and rat [34], these data indicate that translational efficiency of GS mRNA in both species is similar postnatally. Applying a similar reasoning to the observed

time constant for GS accumulation before birth (0.003-day^{-1}) indicates that GS protein does not degrade appreciably before birth and that its rate of synthesis is $\approx 0.05\text{ U}\cdot\text{g}^{-1}\cdot\text{day}^{-1}$, that is, sixfold lower than after birth. As GS expression is already confined to the pericentral hepatocytes in spiny mouse fetuses [14], these data are consistent with a rather abrupt sixfold perinatal increase in the translational efficiency of GS mRNA. Previously, we reached a similar conclusion for the increase in protein/mRNA ratio for carbamoylphosphate synthetase in rat liver [35,36].

Changes in translational efficiency

An increase in translational efficiency can result from an increase in the availability of mRNA for protein synthesis, that is, from mobilisation of mRNA from ribonucleoprotein particles into polysomes [29], from a change in the rate of initiation of translation, or from a change in the rate of polypeptide chain elongation/termination. If the mRNAs are sequestered from the ribosomes, they are usually found in ribonucleoprotein particles [29]. If initiation of translation is a control point, the polysomes will be sparsely loaded with ribosomes, whereas control at the level of elongation or termination will induce queuing of the ribosomes on the mRNA, that is, heavy loading of the mRNAs. A fully loaded mRNA contains on average one ribosome per 90 nucleotides of the ORF [30]. To determine the loading of an mRNA with ribosomes with reasonable accuracy, it should be relatively short. This was the reason to prefer analysis of GS (ORF: 1122 nt) over glutamate dehydrogenase (ORF: 1674 nt) or carbamoylphosphate synthetase (ORF: 4500 nt). Both in prenatal and postnatal rat [34] and spiny mouse liver (Fig. 4), GS mRNA was found in the polysomal fraction and not in an RNP fraction. Furthermore, the loading of GS mRNAs averages 12 ribosomes and does not change during development. These data also make control at the level of translation initiation unlikely and, instead, point towards elongation as the control point in perinatal changes in translational control.

Changes in the rate of elongation can be detected by determining the ribosome transit time [37]. As it proved impossible in our hands to flood prenatal liver with labelled methionine (no problems were observed in prenatal heart), we have not yet been able to show definitively that the rate of elongation increases perinatally. However, the calculated rates of GS synthesis and degradation allow an approximation. The molecular specific activity of GS in mammals and birds is $\approx 12\text{ U}\cdot\text{mg}^{-1}\text{ protein}$ [34]. Hepatic GS synthesis in postnatal spiny mouse ($0.35\text{ U}\cdot\text{g}^{-1}\cdot\text{day}^{-1}$) therefore approximates $0.7\text{ nmol protein}\cdot\text{g}^{-1}\cdot\text{day}^{-1}$. As postnatal spiny mouse liver contains $\approx 25\text{ fmol GS mRNA}\cdot\text{g}^{-1}$, the rate of GS synthesis is $\approx 19\text{ GS protein molecules per GS mRNA molecule per min}$. With a loading of GS mRNAs with about 12 ribosomes, the ribosome-transit time should equal 40 s. This calculated elongation rate compares well with that actually observed in adult liver (90 s; F. J. van Hemert, M. J. B. Van den Hoff, A. F. M. Moorman & W. H. Lamers, unpublished data) and resembles that found in mammalian cells with stimulated protein synthesis [37]. This rapid rate of elongation, in conjunction with the relative stability of the GS protein, may explain why GS levels in pericentral hepatocytes can accumulate to the very high level of almost 0.2 mM [34]. The deduced ribosome-transit time in fetal hepatocytes, on the other hand, approximates that found in quiescent cells. These calculations therefore support the conclusion that the kinetics of postnatal GS protein accumulation are consistent with an approximately sixfold increase in the rate of polypeptide elongation.

Conclusion

The comparison of GS accumulation in the liver of a precocial species (spiny mouse) with that of an altricial species (rat) has shown that the observed perinatal increases in the protein/mRNA ratio of ammonia-metabolising enzymes should be considered as adaptive features to postnatal life and not as features that reflect the ongoing development of the animals.

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